Waste Right

Household waste collection, starting at ₹10

Scan your waste.
Book a pickup.
Know what happens next.



Zenith Hackathon / 24 October 2026

Everyday waste takes too much effort

The household has to identify it, sort it and arrange collection.

Unclear materials

A food container can include plastic, paper and leftover food.

Too many steps

Taking a photo, filling forms and finding collection help adds friction.

Little visibility

Residents and helpers need a clear record of weight, charges and payment.

A scan that leads to a real pickup

Waste Right connects the resident, the helper and Operations.

One collection record

The resident confirms the items. The helper verifies the material and weight. Operations records the sale, payments and any dispute.

AI suggests. People confirm.

The resident journey

A short flow, from the camera to the collection receipt.

01 Scan and confirm Use the phone camera or a description. Correct the suggested items.

02 Schedule Choose the address, date and time. Check the automatic fee.

03 Review the receipt The helper weighs the waste. Accept the receipt or report a problem.

The user keeps the final say

A camera makes starting easier. Confirmation makes the record trustworthy.

What the app does

Suggests items and materials.
Reads visible labels with OCR.
Asks for clarification when needed.

What the resident controls

Removes a mistaken item.
Corrects mixed materials.
Confirms what will be collected.

Photographs do not establish weight or safe handling.

One mixed-waste example

Illustrative journey, not a recorded customer collection.

Camera view	Fruit scraps and a plastic food container appear together.
Resident review	The resident confirms both and keeps the materials separate if possible.
Helper handover	The helper checks the material, records scale readings and sends the receipt.

The app does not automatically treat the whole load as organic.

Where the collected waste goes

The pilot needs agreed handovers with local processing partners.

Organic waste	Suitable clean organics go to an approved composting or biogas partner. The processor confirms what it accepts.
Dry materials	Paper, plastic, glass and metal go to suitable recyclers after material checks.
Special waste	Batteries, hazardous items and unsuitable loads need a separate specialist arrangement.

Recovery income and programme benefits require actual acceptance and payment records.

Plans for different collection needs

Current pilot prices. One-month passes have no automatic renewal.

	Basic	Gold	Platinum
Price	₹10 / pickup	₹399 / month	₹469 / month
Included pickups	1	14	24
Per pickup limit	2 kg	8 kg	10 kg
Monthly allowance	Pay per pickup	112 kg	240 kg
Segregation	₹5/kg if mixed	Included	Included
Resident recovery share	0%	10% of net	15% of net

Metal counts toward limits. Specialist or hazardous waste needs separate handling.

Clear charges before and after weighing

Residents never have to guess the weight.

Standard hours	7 am to before 2 pm. No collection from 2 pm to 5 pm.
Evening hours	5 pm to 7 pm. A ₹5–₹7 surcharge depends on time, weight and sorting.
Mixed Basic waste	₹5 per measured kg for segregation. Gold and Platinum include segregation.

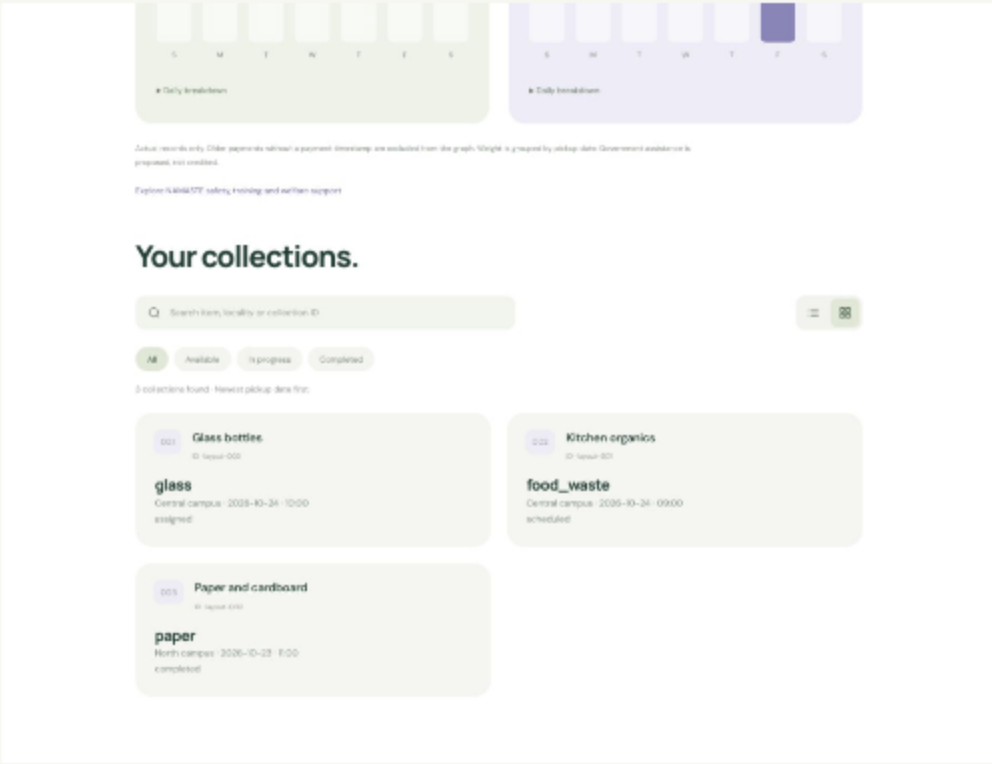
Example: 2 kg of mixed Basic waste at 10 am = ₹10 pickup + ₹10 sorting = ₹20.

Helpers get a dedicated workspace

English and Hindi support the collection workflow.

Collections have their own page

Search by item, locality or ID.
Switch between grid and list.
Accept jobs and record weight.
Track receipts and status.



Synthetic layout example

Helper earnings are visible

The pilot separates earned pay, recorded payments and outstanding amounts.

₹8

per completed pickup

+ 50% of sorting service value

+ 40% of net recovery

Earnings require a completed, accepted receipt. Operations records actual payments.

Operations knows what to do next

A separate workspace for Collections, Payments, Helpers and Issues.

A collection queue

Search records and open one collection at a time.

Clear payment steps

See earned, paid and outstanding helper amounts. Record a completed transfer.

Separate sale records

Record actual recycler income and costs, then the resident’s recovery share.

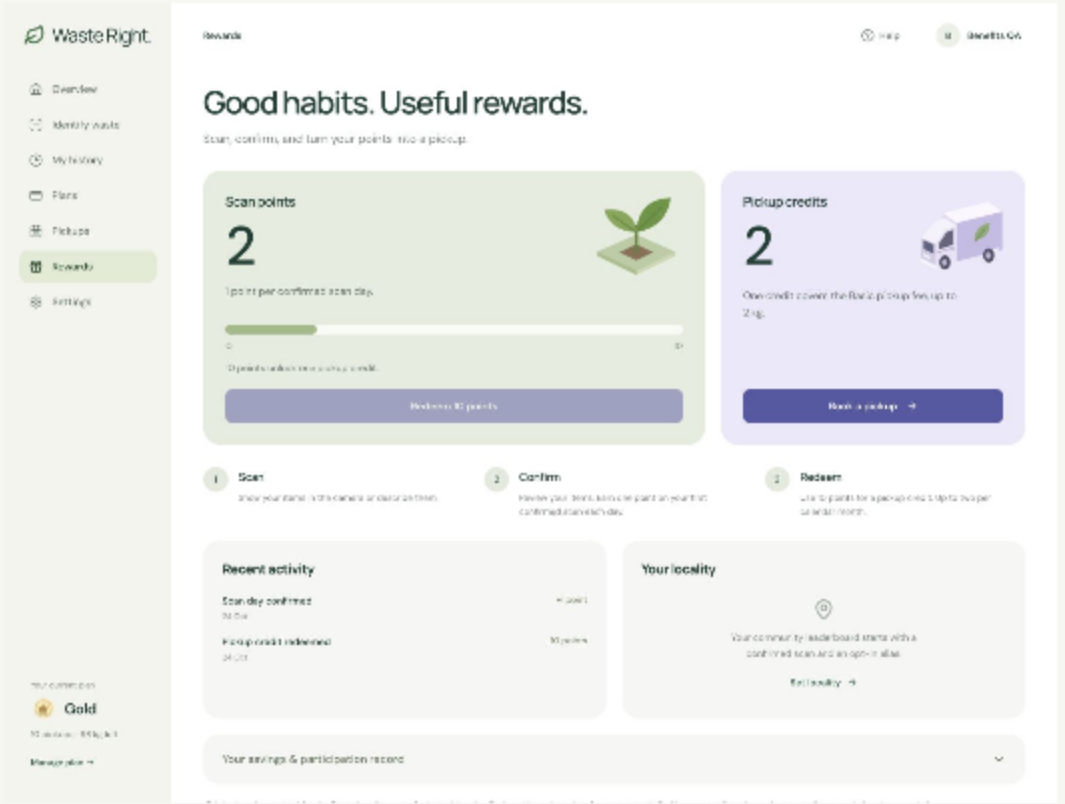
Rewards make the habit easier to keep

Earn one point on the first confirmed scan day.

10 points

unlock one pickup credit

Up to two credits per calendar month.
Each covers the Basic fee up to 2 kg.



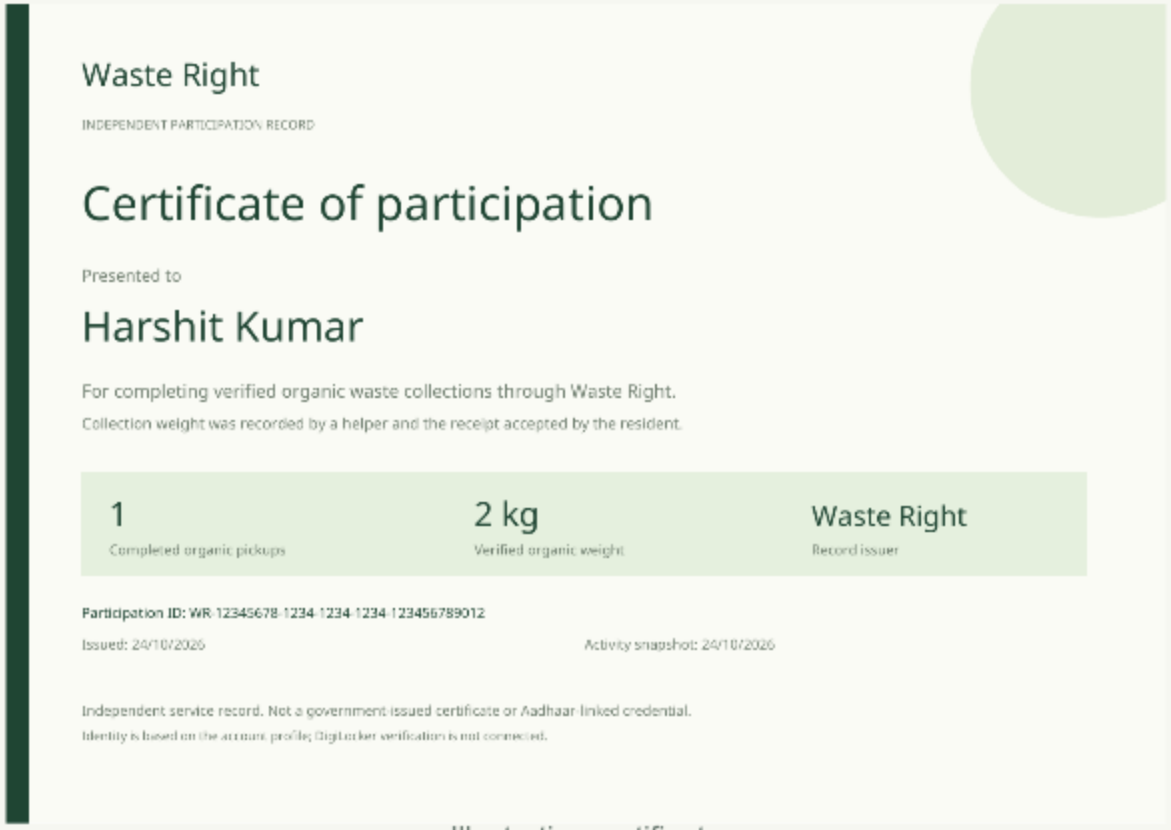
A useful record of participation

A completed organic pickup unlocks a downloadable Waste Right certificate.

Name and participation ID

Completed organic pickups
Helper-measured organic kg
Issue date and activity snapshot

Independent issuer: Waste Right



Illustrative certificate

How Waste Right earns revenue

Customer payments and verified material recovery fund the service.

Pickup fees	Basic collections generate a fee for each completed booking.
Monthly passes	Gold and Platinum generate revenue when residents buy or renew their pass.
Material recovery	Accepted recycler or processor proceeds contribute after direct costs and shares.

Government support adds income only after eligibility and actual receipt.

A ₹10 pickup leaves a narrow margin

Illustrative base-fee calculation before travel and other costs.

₹10

customer pickup fee

₹8

base helper pay

₹2

left before other costs

Dense routes, subscriptions and actual recovery must cover transport, sorting, payment fees and overhead.

Government support: a defined future channel

Approval is a model assumption. Current scheme terms and funding need confirmation.

50%

of attributable support received goes to the resident as a recorded credit

Support follows eligible output

Published MDA: ₹1,500 per tonne of eligible organic fertiliser.

Household pickup weight cannot directly determine this payment.

A partnership, grant, PPP status or government credential is not claimed as secured.

Support for collection partners

Safe work and access to welfare matter alongside pay.

Our service responsibilities

Clear pay records and dispute handling.
Review route time and travel costs.
Plan safety equipment and training.

Potential NAMASTE support

Validated waste pickers may qualify for PPE, safety training and health coverage under scheme rules.

Waste Right signup alone does not establish government eligibility.

AI assists the collection decision

The current app combines image understanding, OCR and resident review.

Image understanding	Suggests likely materials from camera frames or a description.
OCR	Reads visible packaging text to add context.
Human confirmation	The resident confirms the list. The helper establishes actual material and weight.

Our own organic-waste model

A trained research model separates organic regions from visible contaminants.

450

labelled source images

67

held-out test images

66.13%

foreground mean IoU

IoU measures overlap between predicted and labelled regions. It is not item-identification accuracy.

One household and one fixed camera. Phone-camera validation is still needed.

A working app with a separate model service

Simple components keep the demo understandable and the model replaceable.

Web app	Next.js and React with role-based resident, helper and admin workspaces.
Application records	Local MongoDB stores accounts, bookings, receipts, benefits and payment records.
AI and payments	Configurable image AI and OCR. Razorpay checkout integration. Separate local Python inference API.

What works today

A tested prototype, with clear limits before launch.

Built in the prototype

Camera and description flow
Automatic fees and scheduling
Helper collections and earnings
Receipts, disputes and PDF certificates

Before a public launch

Live payment and bank checks
Field tests on phones and routes
Approved processor agreements
Government eligibility and identity onboarding

31 tests passed, plus isolated API integration checks and a production build.

A focused first pilot

Proposed scope: one campus or housing cluster, with one processing partner.

Start small	Target 50 households and two approved helpers for a four-week pilot.
Measure the work	Track completed pickups, route minutes, kg accepted, repeat bookings and disputes.
Measure the money	Track helper hourly take-home and contribution after every direct cost.

Pilot gate: reliable collections, positive route contribution and acceptable helper earnings.

What we need next

A pilot location and a processing partner to test the complete service.

A campus or housing association

A local waste processor and collection partners

Support to measure real costs and improve the service

Appendix: rewards and service rules

Useful details for reading the deck without a presenter.

Points and credits

One point per first confirmed scan day in India. Ten points per credit. Maximum two credits per calendar month.

Savings and history

Savings separate fee reductions, included segregation and recorded government credits. History stores scans, receipts and disputes.

Certificate and identity

Completed measured organic collections unlock the PDF. Waste Right issues it using the account name. DigiLocker and Aadhaar linking remain unconfigured.

Appendix: pricing and earnings examples

Illustrative calculations under the current pilot policy.

2 kg mixed Basic pickup	Customer pays ₹20 in standard hours. Helper earns ₹8 + ₹5 sorting share = ₹13 before any recovery share.
₹100 positive recovery pool	Helper recovery share is ₹40. Gold resident share is ₹10, or Platinum ₹15. Remaining ₹50 or ₹45 is before other business expenses.
₹100 attributable support received	Resident credit policy allocates ₹50 after completed processing, receipt and credit records. No projected grant appears as savings.

Sources and evidence

The pitch uses project records and official scheme information.

Product and business rules

Current app catalog and pricing logic
Helper pay and benefit accounting
Certificate eligibility and edge-case audit

Model evidence

Household-v2 results and model card
TU Wien dataset, CC BY 4.0
doi.org/10.48436/27k90-dvw73

Government sources: PIB, 10 March 2026 (MDA); Department of Social Justice (NAMASTE); DigiLocker partner portal.